IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected Copy

COMMUNICATION NETWORKS

Thursday, 10 December 9:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : J.A. Barria
Second Marker(s) : J.V. Pitt

© Imperial College London

Special information for students

1. Mean delay for the M/M/1 system may be taken as

$$t_i = \frac{1}{\mu C_i - \lambda_i}$$

where,

$1/\mu$ = Average length of packet [bits / packet]

C_i = Transmission speed link i [bits / s]

μC_i = Service rate (link i) [packets / s]

λ_i = Arrival rate (link i) [packets / s]

2. Optimal Routing Problem (ORP)

$Min\ D(x)$ with respect to $x = \{ x_i \}$

where $D(x) = \sum_{i=1}^{L} \frac{x_i}{C(i) - x_i}$

$C(i)$ = capacity of link i,

x_i = flow carried by link i,

L = total number of links in the network.

The Questions

1.

a) For a Stop-and-Wait ARQ protocol, the time to send a frame and receive an ACK (acknowledgement) frame, in the absence of errors, is given by:

$$t_0 = 2t_{prop} + 2t_{proc} + t_f + t_{ack} = 2t_{prop} + 2t_{proc} + \frac{n_f}{R} + \frac{n_a}{R}$$

where, t_{prop} = is the propagation delay; t_{proc} = is the processing delay; n_f = is the number of bits in the information frame; n_a = is the number of bits in the ACK frame, and R = is the bit rate of the transmission channel.

The effective information transmission rate of the protocol in the absence of errors is given by:

$$R_{eff}^0 = \frac{n_f - n_O}{t_0}$$

where, n_O = is the number of overhead bits in a frame (header plus CRC check bits). Define the transmission efficiency η_0 of this protocol as:

$$\eta_0 = \frac{R_{eff}^0}{R}$$

i) Derive η_0 as a function of all the protocol parameters appearing in the above expression for t_0. [3]

ii) For the expression derived in i) clearly identify and explain all the sources of inefficiency. [6]

b)

i) Explain and discuss why it important to implement a collision avoidance (CA) mechanism in a wireless communication environment. [3]

ii) The protocol 802.11 has an optional feature that enables the implementation of a reservation scheme.

- Explain why a reservation protocol would be needed. [2]
- Discuss under which conditions it would be useful to implement this optional feature. [2]
- Sketch the time sequence of the 802.11 reservation protocol. [4]

2.

a)

i) Describe a mechanism that would allow routers to distinguish between different classes of traffic in a multimedia IP environment.

[4]

ii) Describe a mechanism that will provide a degree of isolation among different traffic classes, so that one packet flow is not adversely affected by another misbehaving flow.

[4]

iii) Describe and discuss a mechanism known to you that needs to be in place in the event that there are not enough resources in the network to support the level of service requested by a new call request.

[4]

b) An optimal routing problem aims at allocating flows x_l to all the links l of a network, such that an objective function $D(C,X)$, which represents the network congestion level, is minimised with respect to $X = \{x_l\}$.

Note: in function $D(C,X)$:
$C = \{C(i)\}$ is the set of capacities of links l, and
$X = \{x_l\}$ is the set of flows carried by links l.

i) Using the results of a single $M/M/1$ queue system, derive an objective function $D(C,X)$ that would represent the mean network delay incurred by packets traversing a network of $M/M/1$ queues. Clearly show all the steps of your derivation.

[4]

ii) Discuss and suggest an additional term to function $D(C,X)$ that will have the effect of controlling the flow of packets that is carried by the network.

[4]

3.

a) In the context of IP mobility support using indirect routing:

i) Describe and discuss the functionalities required of the following entities:

- Foreign agent, [4]

- Home agent. [4]

ii) Explain and discuss why indirect routing requires an encapsulation / decapsulation function. [4]

b) The goal of a statistical shortest path problem is to minimise the length L_p of path p:

$$L_p = \Psi(\mu_p) + \Phi(\sigma_p^2)$$

where μ_p is the mean value of L_p; σ_p^2 is the variance of L_p and, $\Psi(.)$ and $\Phi(.)$ are arbitrary functions.

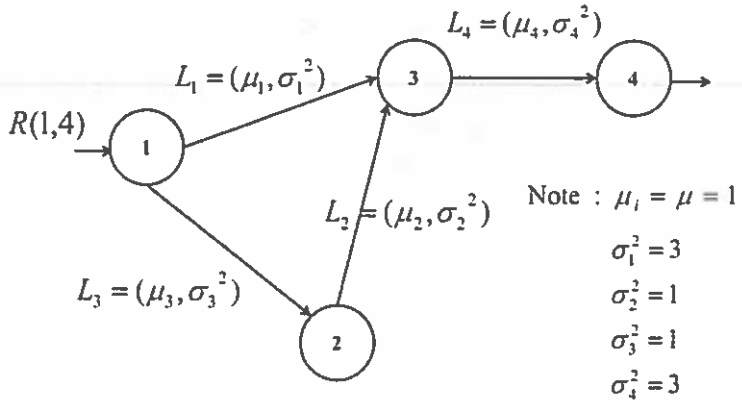
i) Assume that the links' statistics of a network are Gaussian distributed, with mean μ_i and variance σ_i^2 for link i, and that the links in the network are mutually independent.

- For the network shown in Fig. 3.1, derive an expression of the form $L_p = \mu_p + \sigma_p^2$ for all identifiable paths p. Discuss your answer. [2]

- Using the network and information shown in Fig. 3.1 identify the shortest path in terms of σ_p^2. Explain the logic of your derivation. [3]

ii) Now consider a modified cost function with $\Phi(x) = \sqrt{x}$.

- Is it possible to use the same approximation used in i) to find the shortest path in respect to $\sqrt{\sigma^2} = \sigma$? Discuss your answer.



[3]

Figure 3.1

4.

a) In the context of multimedia networking, a FIFO (first in first out) queue system requires modifications so that different classes of packets are treated differently.

i) Discuss an alternative way of treating packets arriving to a queue such that arrivals could be receive differential treatment. Describe how you would modify a FIFO queue system to handle differential treatment between two classes of traffic. [4]

ii) Describe what is a weighted fair queueing (WFQ) discipline, and explain how you would implement it. [4]

b) For a TCP dynamic congestion window flow control the following closed-form expression for the latency has been proposed:

$$Latency = 2RTT + \frac{O}{R} + P\left[RTT + \frac{S}{R}\right] - \left(2^P - 1\right)\frac{S}{R}$$

where $P = \min\{Q, K-1\}$; $RTT =$ round trip time; $O =$ size of the object; $R =$ transmission rate of the link; $S =$ the size of the MSS (maximum segment size); $K =$ number of windows that covers the object; $W =$ window size; and $Q =$ is the number of times the server would stall if the object contained an infinite number of segments.

i) Show that a bound on the ratio of Latency to the Minimum Latency (Latency assuming that there is no congestion control), namely its efficiency, is given by: [4]

$$\eta_{eff} = \frac{\text{Latency}}{\text{Minimum Latency}} \le 1 + \frac{P}{[(O/R)/RTT]+2}$$

Hint: First determine the Minimum Latency expression.

ii) Discuss the impact of the magnitude of values of the parameters RTT and O/R on the efficiency of the slow start mechanism. [4]

Question 4 continues next page

Question 4 - b) iii)

iii) For the following parameters: $S = 537$ bytes, $RTT = 100$ ms, $O = 100$ Kbytes and $K = 8$, the following calculations are given:

R (transmission rate)	Minimum Latency	Latency - SLOW START
Kbps	s	s
28.0	28.8	28.9
100.0	8.2	8.4
10.0	0.28	0.98

- What can you say about the effects of the transmission rate R on the efficiency, η_{eff}, of the slow start? [4]